

Milestone 1 — Exploratory Data Analysis (EDA) Report

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Dataset: Bank Customer Churn Prediction (Kaggle) | **Deliverables:** Hloz_M1.ipynb, Hloz_M1_Cleaned.csv, Hloz_M1_Report.pdf

1. Dataset and Target

Dataset Purpose and Link: The dataset used in this milestone is the *Bank Customer Churn Prediction* from Kaggle. The goal is to understand what customer features are related to customers leaving the bank (churning). Kaggle link:

<https://www.kaggle.com/datasets/shantanudhakadd/bank-customer-churn-prediction>

What One Row Represents: Each row in the dataset represents a single bank customer, containing their personal information (such as country, gender, age), account numbers (credit score, balance, number of products), activity status, and whether they stayed or left.

Number of Rows and Columns: The raw dataset has **10,000 rows (samples)** and **14 columns (features)**.

Target Column Name and Classes: The target column is **Exited**. It tells us whether the customer stayed or churned. It is a binary classification problem with 2 classes:

Target Class	Meaning	Count (Rows)	Percentage (%)
0	Customer stayed (Retained)	7,963	79.63%
1	Customer left (Exited / Churned)	2,037	20.37%
Total	All Customers	10,000	100.00%

Input Features Classification:

- **ID-like Columns (To Drop):** RowNumber (row counter), CustomerId (unique ID), and Surname (customer name). These are unique identifiers that do not provide general learning signals, so we drop them.
- **Categorical Input Features (2):** Geography (France, Spain, Germany) and Gender (Female, Male).
- **Numerical Input Features (8):** CreditScore, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, and EstimatedSalary.

2. Initial Inspection and Data Quality

- **Duplicate Rows:** We checked for duplicate rows using `df.duplicated().sum()`. There are **0 duplicate rows** in the dataset.
- **Missing Values:** We counted missing values across all columns. There are **0 missing values** in the entire dataset (100% complete).
- **Class Balance:** The target is moderately imbalanced (roughly 80% stayed vs 20% left). In future milestones, we should check precision, recall, and F1-score rather than just accuracy.
- **Unusual Values:** Over 36% of bank customers have an account balance of exactly 0. This creates a huge spike at 0 in the balance distribution.

3. Descriptive Statistics for Numerical Features

Here is the summary table for the 8 numerical input features (\$N = 10,000\$), showing count, mean, standard deviation, minimum, median, and maximum.

Feature	Count	Mean	Std Dev	Min	Median	Max	Shape
CreditScore	10,000	650.53	96.65	350.00	652.00	850.00	Bell-shaped
Age	10,000	38.92	10.49	18.00	37.00	92.00	Right-skewed
Tenure	10,000	5.01	2.89	0.00	5.00	10.00	Uniform (0-10)
Balance	10,000	76,485.89	62,397.41	0.00	97,198.54	250,898.09	Bimodal (many 0s)
NumOfProducts	10,000	1.53	0.58	1.00	1.00	4.00	Discrete (1-4)
HasCrCard	10,000	0.71	0.46	0.00	1.00	1.00	Binary (70.6% 1)
IsActiveMember	10,000	0.52	0.50	0.00	1.00	1.00	Binary (51.5% 1)
EstimatedSalary	10,000	100,090.24	57,510.50	11.58	100,193.92	199,992.48	Uniform

4. Target Distribution and Missing Values Plots

Below are the plots showing the target class distribution and the missing values check before preprocessing.

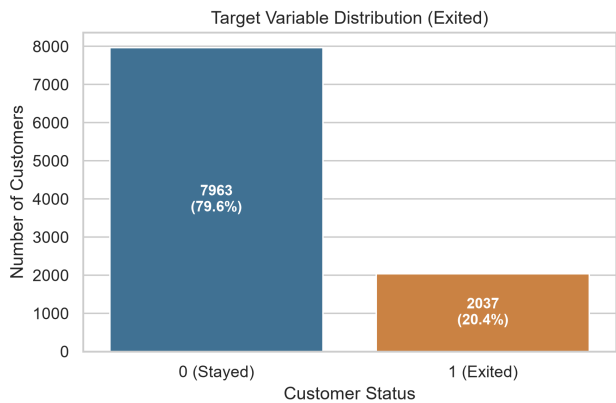


Figure 1: Number of samples in each target class (Exited).

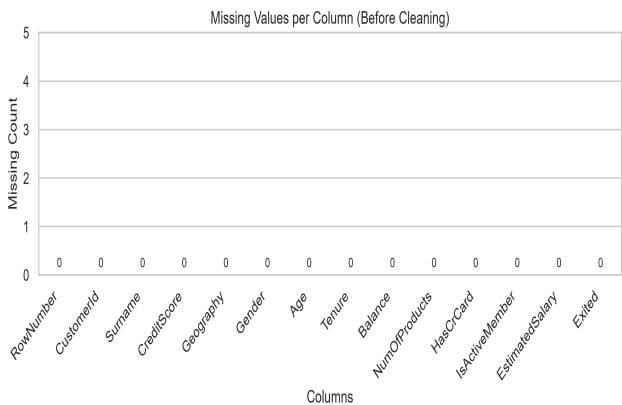


Figure 2: Missing values count for all columns before cleaning.

Observations:

- **Target Distribution:** 7,963 customers (79.6%) stayed, and 2,037 customers (20.4%) left. The target is imbalanced with a 4:1 ratio.
- **Missing Values:** Every single column has 0 missing values. The dataset is fully complete.

5. Estimated PDF for Numerical Features

Here we plot the estimated Probability Density Function (PDF) using normalized histograms (density=True) with a KDE curve for all 8 numerical features.

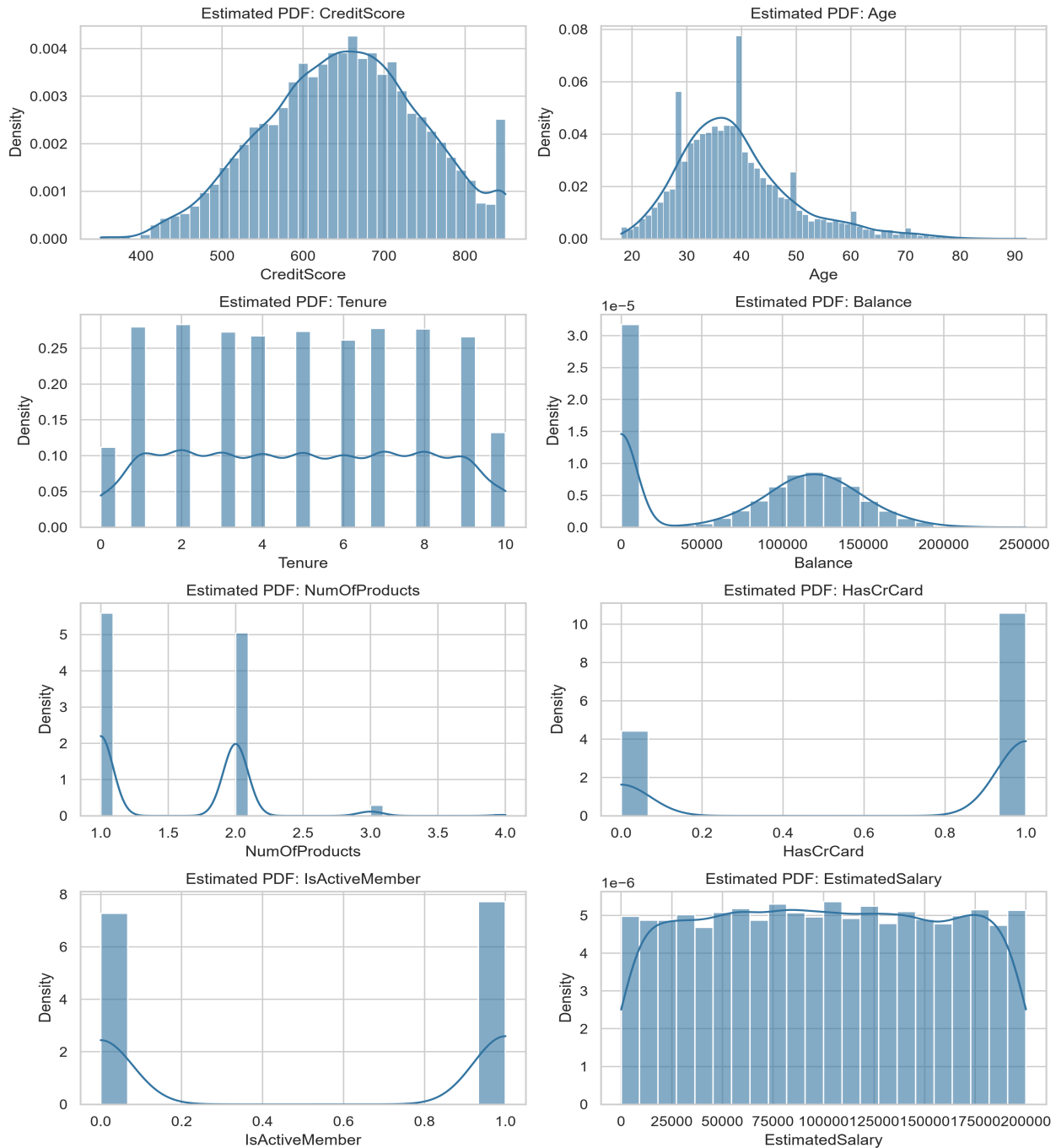


Figure 3: Estimated PDF plots with KDE curves for all 8 numerical features.

Observation: CreditScore has a symmetric, bell-shaped distribution centered around 650. Age is skewed to the right, with most customers between 30 and 45 years old. Balance has a large spike at zero (customers with zero balance) and another bell shape around 120,000 for customers with money in the bank. EstimatedSalary is flat across the whole range from 0 to 200,000, looking like a uniform distribution.

6. Empirical CDF for Numerical Features

Below are the Empirical Cumulative Distribution Function (CDF) step plots for all 8 numerical features, showing cumulative probability $F(x)$.

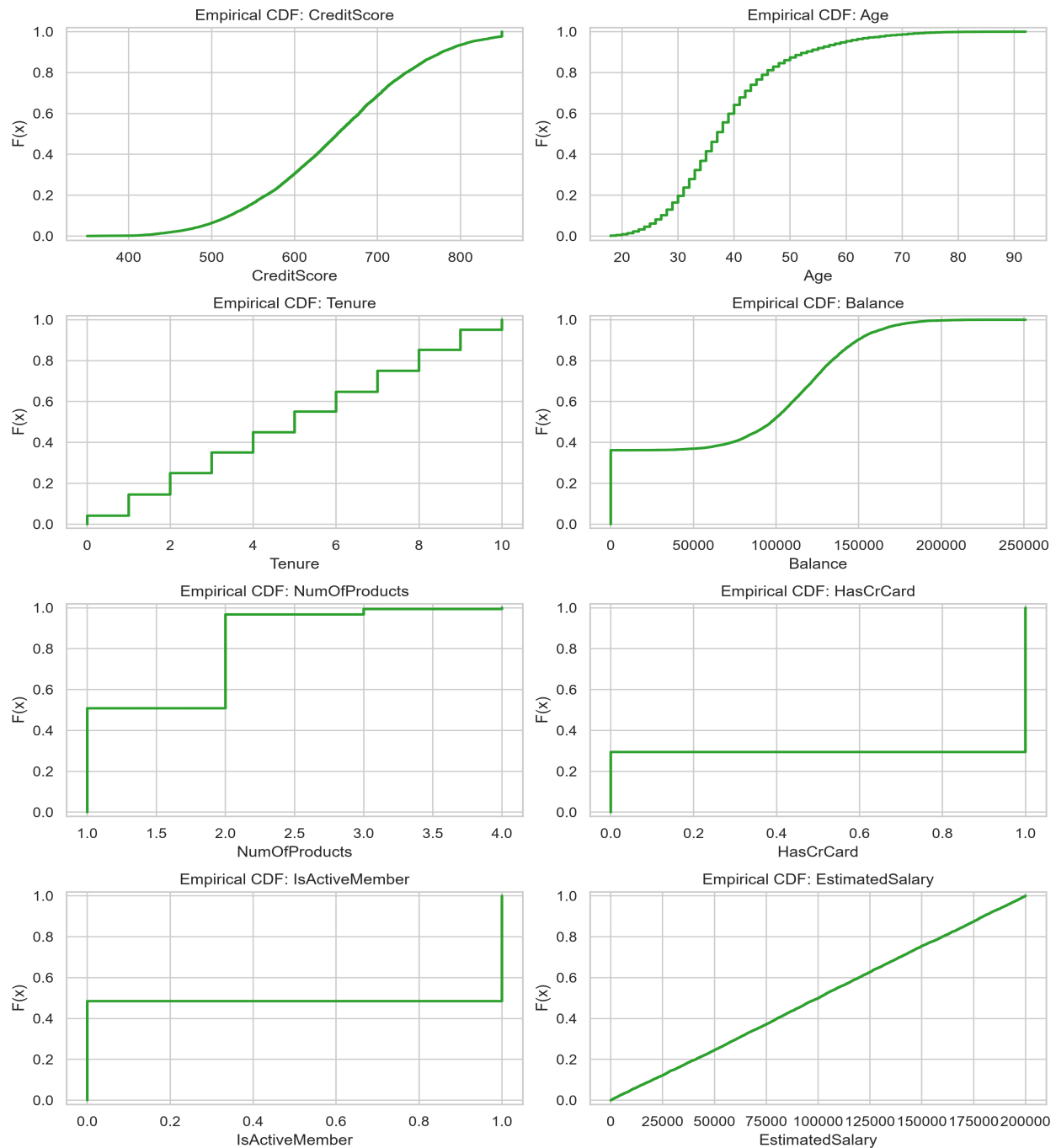


Figure 4: Empirical CDF step plots for all 8 numerical features.

Observation: The CDF for Balance jumps vertically at 0 up to about 0.36, which confirms that roughly 36% of customers have zero balance. The CDF of EstimatedSalary is a straight diagonal line from (0, 0) to (200000, 1), which is the expected shape of a uniform distribution. The CDFs of CreditScore and Age have smooth S-curves.

7. Correlation Matrix and Heatmap

Figure 5 shows the correlation heatmap between numerical features after preprocessing. Figure 6 shows boxplots comparing Age and CreditScore between customers who stayed vs left.

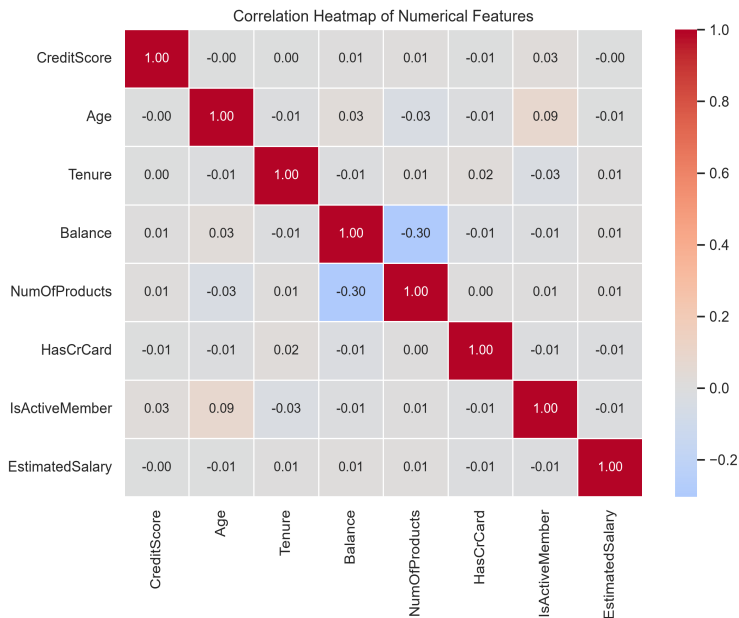


Figure 5: Correlation heatmap of numerical features.

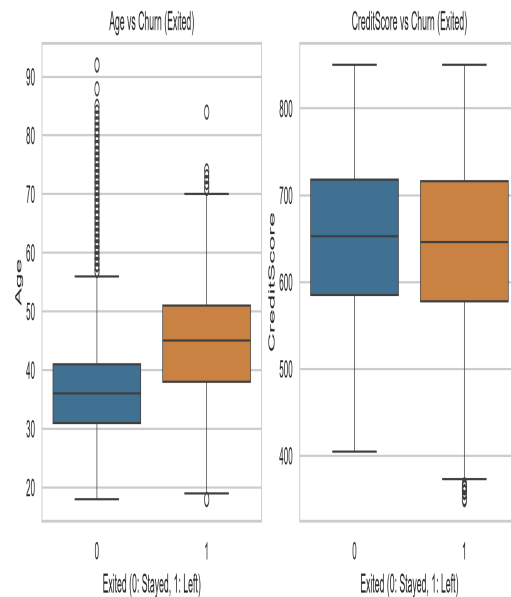


Figure 6: Boxplots of Age and CreditScore vs Exited.

Observation: Almost all numerical features have correlations close to 0. The only small correlation is between Balance and NumOfProducts (-0.30), meaning customers with more products tend to have lower account balances. From Figure 6, older customers are much more likely to exit (churned customers have a higher median age of ~45 compared to ~36 for retained customers). On the other hand, credit scores are almost identical between both groups.

8. Basic Preprocessing Summary

We followed the 4 required preprocessing steps in strict order:

1. **Remove duplicate rows:** 0 rows removed (no duplicate rows existed).
2. **Remove rows with missing target:** 0 rows removed (all rows have a target label).
3. **Fill missing numerical values with median:** 0 values filled (no null values were present).
4. **Encode categorical features and target:** Dropped ID columns (RowNumber, CustomerId, Surname). One-hot encoded Geography into 3 binary columns (Geography_France, Geography_Germany, Geography_Spain) and Gender into 2 binary columns (Gender_Female, Gender_Male). Target Exited was kept as integer labels (0 and 1).

9. Final Cleaned Dataset Specification

The cleaned dataset was saved to **Hloz_M1_Cleaned.csv**. It has **10,000 rows** and **14 columns**. Here is the column dictionary:

Column Name	Data Type	Transformation	Values / Range
CreditScore	Integer	Raw numerical	[350, 850]
Age	Integer	Raw numerical	[18, 92]
Tenure	Integer	Raw numerical	[0, 10]
Balance	Float	Raw numerical	[0.00, 250,898.09]
NumOfProducts	Integer	Raw numerical	[1, 4]
HasCrCard	Integer	Binary indicator	{0, 1}
IsActiveMember	Integer	Binary indicator	{0, 1}
EstimatedSalary	Float	Raw numerical	[11.58, 199,992.48]
Exited	Integer	Target (0: Stayed, 1: Left)	{0, 1}
Geography_France	Integer	One-Hot Encoded from Geography	{0, 1}
Geography_Germany	Integer	One-Hot Encoded from Geography	{0, 1}
Geography_Spain	Integer	One-Hot Encoded from Geography	{0, 1}
Gender_Female	Integer	One-Hot Encoded from Gender	{0, 1}
Gender_Male	Integer	One-Hot Encoded from Gender	{0, 1}

10. Conclusion and What We Learned (5–8 Lines)

In this milestone, we performed an exploratory data analysis on the bank churn dataset with 10,000 customers. The data is very clean with zero duplicate rows and zero missing values across all features. We observed that the target variable Exited is imbalanced, with roughly 79.6% of customers staying and 20.4% leaving. Distribution plots showed that CreditScore is normal, EstimatedSalary is uniform, and Balance has a large group of customers with zero balance. Our analysis showed that age is strongly related to churn, with older customers leaving more frequently than younger ones. Finally, we removed useless ID columns and one-hot encoded the categorical features, saving a clean dataset of 10,000 rows and 14 columns into Hloz_M1_Cleaned.csv.

Milestone 1 Deliverables Summary

- ✓ **Hloz_M1_Report.pdf**: Summary report within the 6-page limit.
- ✓ **Hloz_M1.ipynb**: Executable Jupyter notebook with code, outputs, and written observations.
- ✓ **Hloz_M1_Cleaned.csv**: Final cleaned and encoded dataset ready for Milestone 2.